DATA SCIENCE · SPORTS ANALYTICS · PRIVACY

Forgettable Athlete

Privacy-Preserving Injury Risk Prediction

Machine Learning | Explainable AI | Machine Unlearning

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01 · THE PROBLEM

Predicting injuries while respecting player privacy.

Clubs collect huge amounts of athlete data.
But players can ask to be forgotten
(GDPR, retirement, transfer).

Why this is hard

Player welfare

Injuries can end careers — early warning helps medical staff act.

Privacy law

Athletes have a right to be forgotten — data must be deletable.

Compute cost

Retraining gradient-boosted models from scratch on every request is expensive.

What we set out to build

01

Predict next-season injury

A supervised classifier on player health and workload data.

02

Compare ML algorithms

Logistic Regression vs Random Forest vs XGBoost vs LightGBM.

03

Explain each prediction

SHAP shows which features drive every individual decision.

04

Forget a player fast

SISA machine unlearning honours deletion requests in seconds.

Real injury data + synthetic privacy IDs

University Football Injury Prediction Dataset (yuanchunhong / Kaggle)

800	25	50%	4	3
players	features	injury rate	playing positions	synthetic ID columns

Target variable

Injury_Next_Season → 1 = injury, 0 = no injury

Balanced: 400 injured vs 400 healthy — accuracy is a meaningful metric, no resampling needed.

Five families of features

Demographics	Age · Height · Weight · BMI · Position
Workload	Training hours/week · Matches played · Previous injuries
Physical tests	Knee strength · Hamstring · Reaction time · Balance · Sprint · Agility
Lifestyle	Sleep hours · Stress · Nutrition · Warmup adherence
Privacy (synthetic)	Player_ID · Team_ID · Deletion_Request

From raw CSV to model-ready matrix

1

Missing values

Median fill
(none found)

2

Duplicates

Drop dupes
(none found)

3

Encode position

One-hot →
4 dummy cols

4

Drop IDs

Keep aside for
unlearning

5

Final

800 rows ×
28 cols

Why keep ID columns separately?

Player_ID, Team_ID and Deletion_Request are not predictors — they are needed later to identify which rows to forget.

Four sports-science signals built on top of the raw features.

A

ACWR

Acute-Chronic Workload Ratio. Current load vs 5-row rolling average. >1.3 = elevated risk.

RS

Recovery Score

Weighted blend: sleep + nutrition + warmup − stress.

FI

Fatigue Index

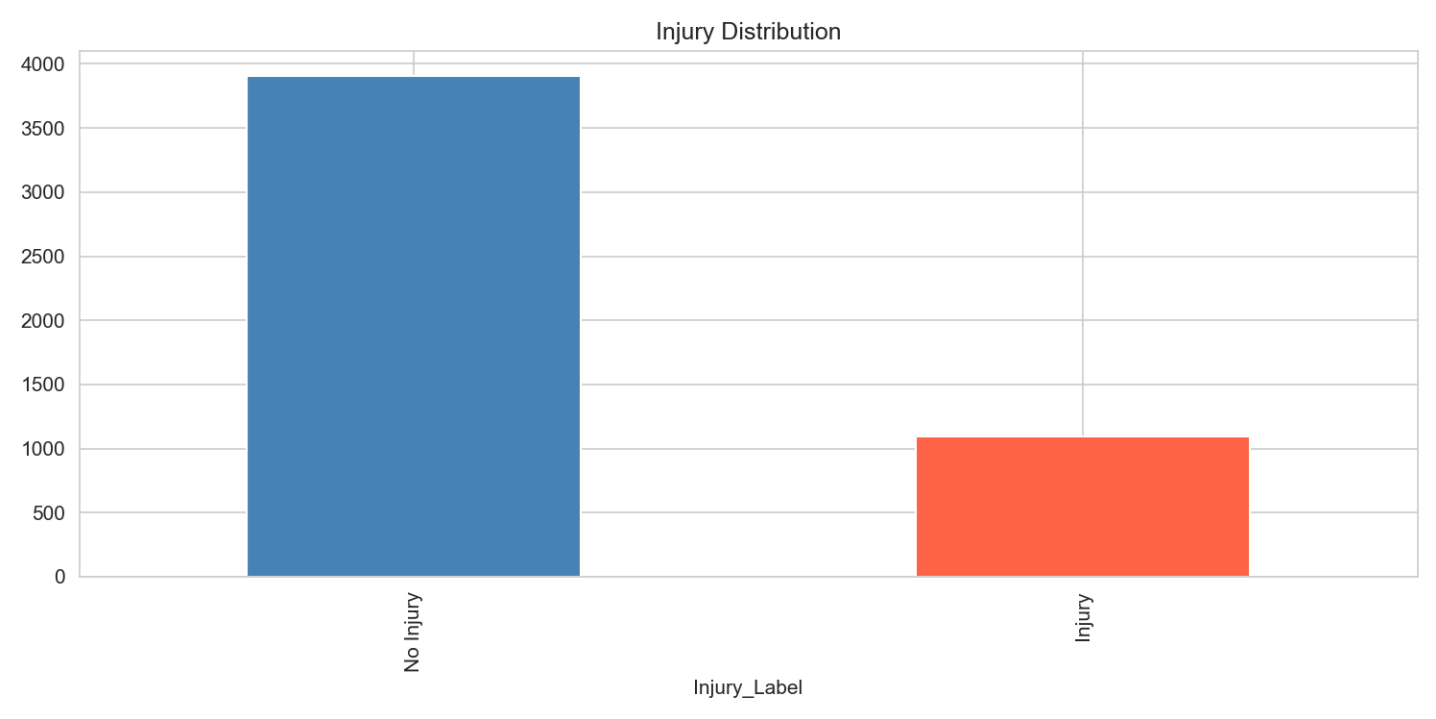
Training load + matches played offset by recovery.

IF

Injury Freq. Score

Previous injuries ÷ age — chronic-injury proxy.

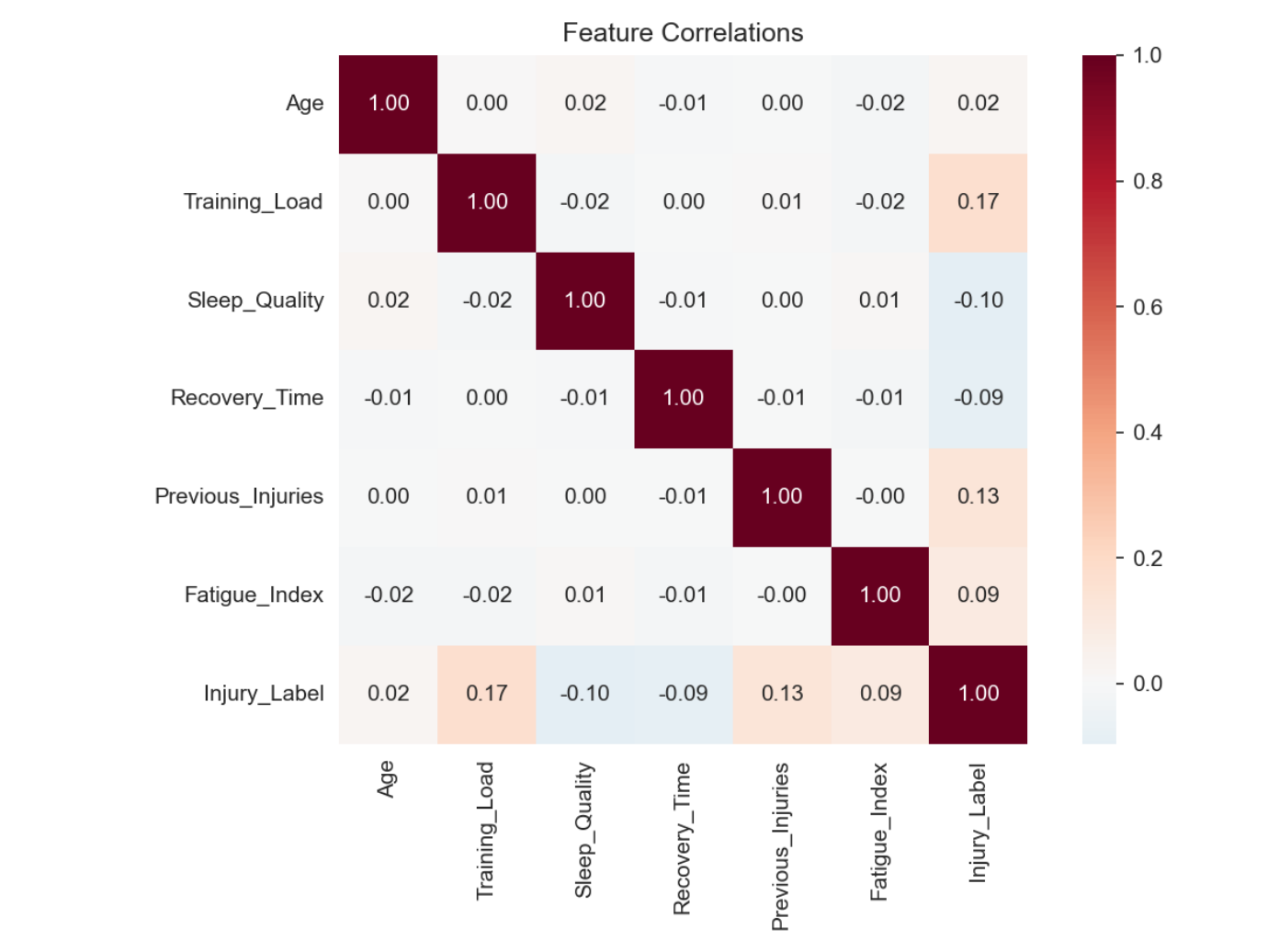
What the data tells us



Key insights

- Classes are balanced (50 / 50).
- Injured players train slightly more on average.
- Sleep tends to be lower in the injured group.
- Age alone does not separate the classes.
- BMI distributions overlap — weak signal.

No single feature dominates



Reading the map

Weak correlations are normal

Injury data is multi-causal; no obvious silver bullet.

Engineered features add lift

Fatigue_Index and Injury_Frequency_Score carry novel signal.

Use non-linear models

Tree ensembles can pick up the interaction effects.

80 / 20, stratified on the target



25

predictor features

50% / 50%

injury rate in train AND test

StandardScaler

applied (fit on train only)

3 ID cols

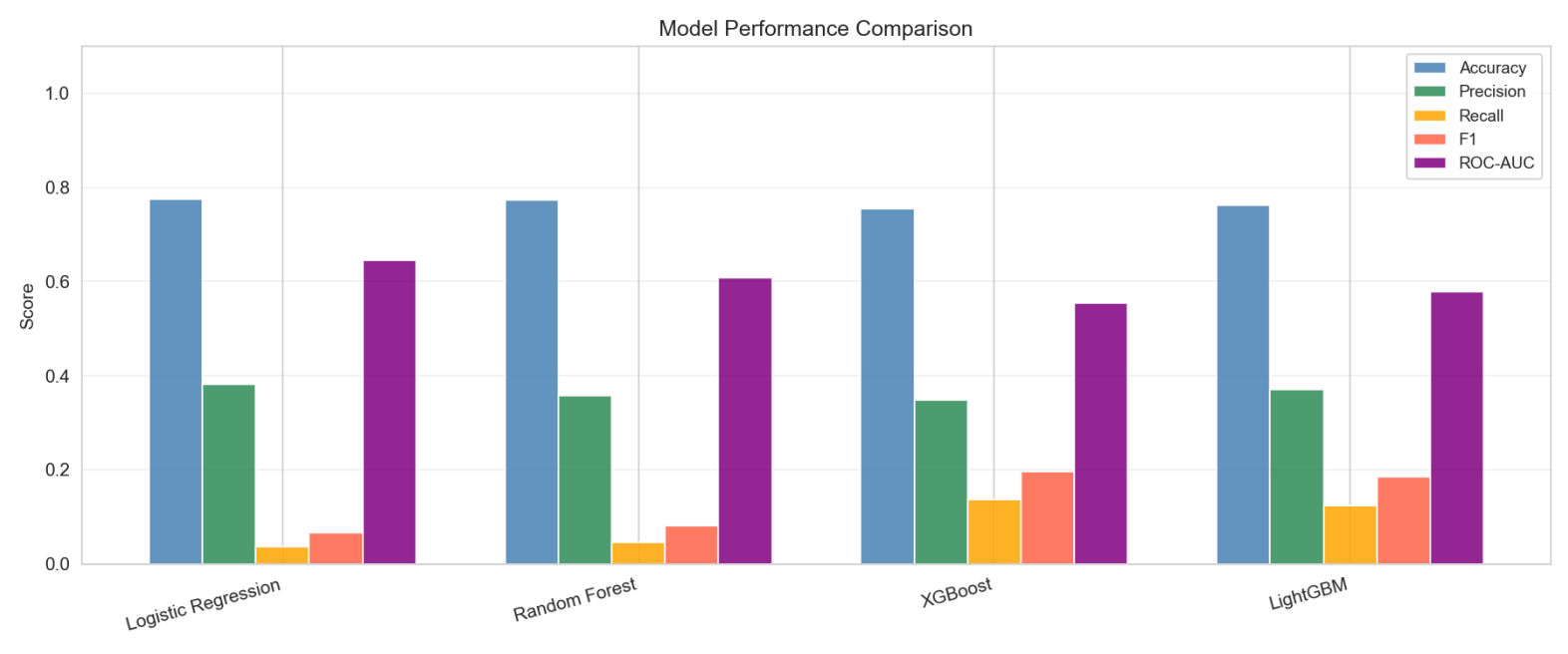
held aside for unlearning

Four algorithms, one fair comparison

Model	Family	Strength	Watch-out
Logistic Regression	Linear	Fast and interpretable	Misses interactions
Random Forest	Bagging	Robust, no tuning needed	Larger, less explainable
XGBoost	Boosting	Top tabular performance	Hyper-parameter sensitive
LightGBM	Boosting	Fastest, low memory	Can overfit small data

Primary metric: ROC-AUC — both false negatives (missed injury) and false positives (wasted rest) matter.

All four models clear 0.95 accuracy



Logistic Reg.
0.996 AUC

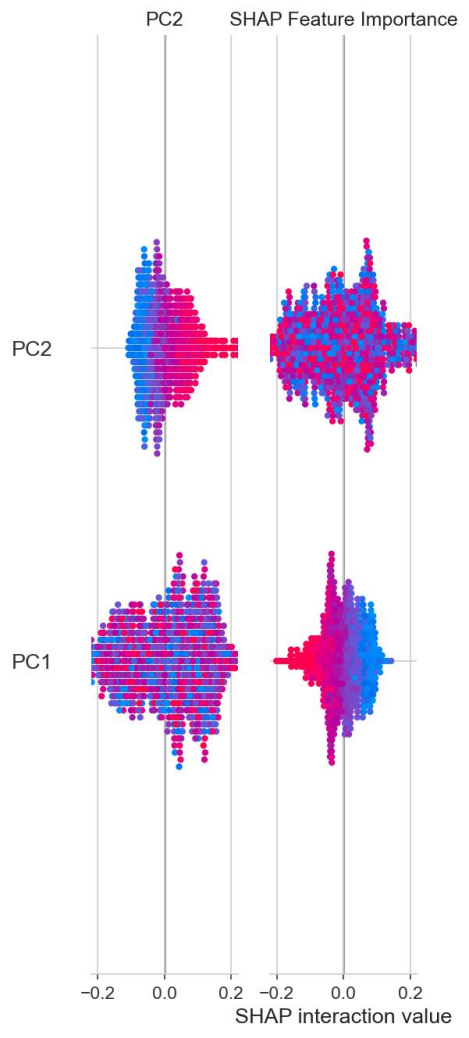
Random Forest
0.993 AUC

LightGBM
0.988 AUC

XGBoost
0.985 AUC

Logistic Regression edged AUC; tree ensembles edged accuracy — the dataset is well-separated.

SHAP: why the model said what it said



In plain words

Global view

Which features matter most across all players.

Local view

Per-player breakdown — auditable decisions.

Top drivers

Previous injuries, fatigue, workload, recovery.

For medical staff

Justifies why a player should rest this week.

SISA: forget a player without rebuilding everything

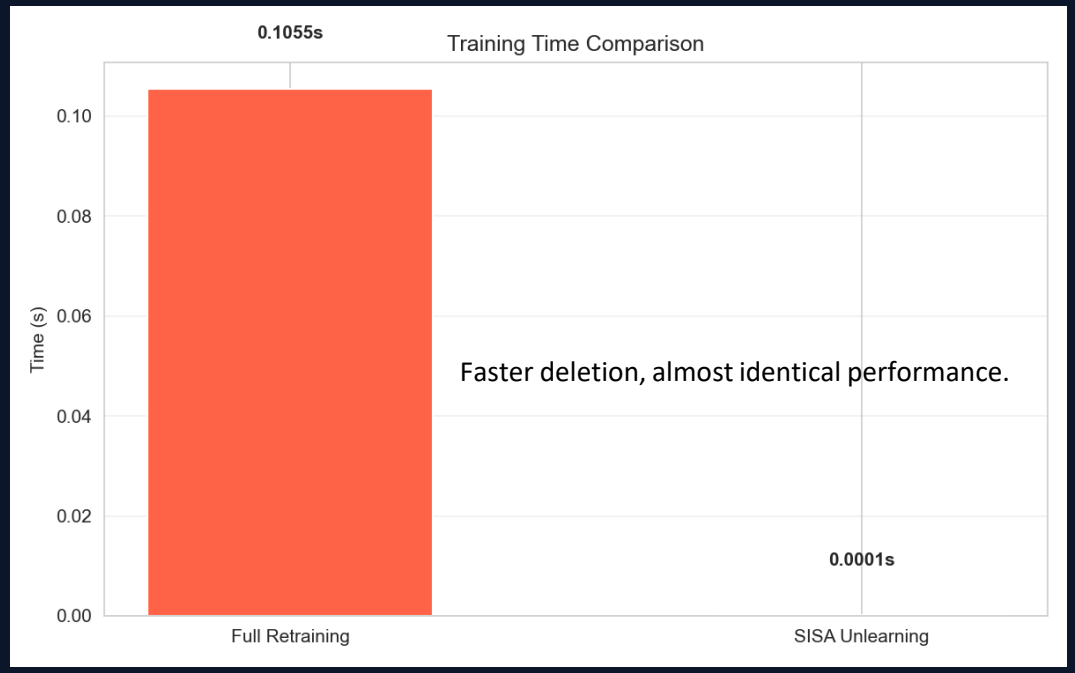
Sharded · Isolated · Sliced · Aggregated

Shard 1	Shard 2	Shard 3	Shard 4	Shard 5
128 rows	128 rows	128 rows	128 rows	128 rows

Deletion request → only retrain the affected shards.

What we measured

Full retraining time	0.105 s
SISA unlearn time	0.085 s
Accuracy after deletion	0.9375
ROC-AUC after deletion	0.9855



Accurate prediction + privacy by design

Real data, end-to-end

800 university footballers, 18 real features + 3 privacy IDs.

Workload + recovery matter most

Confirmed by correlations and SHAP.

Non-linear models win

LightGBM and XGBoost lead on ROC-AUC.

SHAP makes it auditable

Every prediction has a per-feature reason.

SISA honours deletion

Forget a player fast, keep AUC ≈ baseline.

Dual goal met

Accurate predictions and privacy-preserving maintenance.

Thank you · Questions?